

# Stephen Purdue

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## EDUCATION

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**Royal Holloway, University of London**

*Master of Science - Artificial Intelligence*

London, UK

2026 - 2027

**University of Winchester**

*Bachelor of Science - Computer Game Development*

Winchester, UK

2023 - 2026

## WORK EXPERIENCE

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**Sous-Chef, Chef de Partie**

*Mustang Group, Southampton, UK*

2021 - 2026

- Led a kitchen of five through high-volume, deadline-driven service under constant time pressure, requiring rapid problem-solving, attention to detail, and direct communication with other colleagues in an agile environment.
- Mentored and trained 4-8 junior chefs and new starters per intake, maintaining consistent service standards and reducing onboarding time during high staff turnover.
- Maintained stock rotation and portion-control procedures to reduce food waste and control ingredient costs across a high-volume kitchen.

## PROJECTS

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- **Stock Market Prediction Engine**, Developed an LSTM model in PyTorch to forecast daily stock prices from historical time-series analysis data, reducing prediction error by 81% versus a moving-average baseline. Containerised training data with Docker for reproducible runs. [GitHub](#)
- **Adaptive Combat Simulation**, Developed an adaptive combat simulator, achieving a 75% win rate against a baseline agent, with overall reward improving from 32% to 87%, measured over 30 unique evaluation simulations by training a PPO-agent using Python and TensorFlow. [GitHub](#)
- **Procedural Generation Algorithm**, Engineered a scalable procedural generation system in Unity using C# and Binary Space Partitioning, generating room layouts and connections in linear time. Reduced level build-time by 77% by implementing modular, parameter-driven logic. [GitHub](#)
- **Ad Trigger Discovery Engine**, Built the analysis backend for an ad-tech pipeline, generating synthetic user prompts across intent buckets and running BM25/embedding-based retrieval to surface and rank high-intent keyword "triggers" from crawled brand data, with full source traceability for review. [GitHub](#)

## TECHNICAL SKILLS

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**Programming Languages:** Python, C#, C++, SQL

**Frameworks and Tools:** Linux, Git, Docker, CI/CD, PyTorch, TensorFlow

**Cloud:** AWS (S3, SageMaker)

## INTERESTS

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- Snowboarding, Competitive Gaming